



The co-founders (L-R): Govinda Balaji (Engineering), Swapnil Shrivastav (Business), Pardeep Garg (Technology & Business), Venkatesh R (Design & Operations)

The company's initial prototype utilised silica gel and had mirrors to focus solar energy and heat on the silica gel to produce water. The self-contained unit could produce about 5 litres of water a day. Each unit required dedicated components like fans, valves, and pumps, which affected the cost while scaling.

To optimise the size and scale of production, the startup switched to a liquid desiccant—calcium chloride solution. The desiccant is sprinkled through a mesh in the absorber units to increase the surface area

as a fan draws air. After it absorbs moisture from the air, the desiccant is pumped to a separate desorber unit where the solution is heated to 65 to 70 degrees celsius by a coiled pipe filled with hot water. The hot and humid air is passed to a low-power, air-cooled condenser. Each absorber unit can collect as much as 200 litres per day. While the output of traditional condensing systems can vary considerably depending on the local humidity, Uravu's device can achieve the same output in drier areas by simply loading more desic-

cant into the absorber.

The company has coupled six absorbers with two desorber units to create Uravu's 1,000-litre-per-day machine. The latest upgrade to Uravu's design can be coupled with different heat sources, including solar thermal panels, biomass burners, and industrial waste heat.

Based in Bengaluru, the startup's lone design and assembly unit has a total in-house capacity of 3,000 litres per day, out of which currently, 1,800 litres are being utilised daily for customers in the hospitality industry.

It is eyeing the production of other beverage products at the plant, such as tonic water, ginger ale, and flavoured water, by setting up an assembly line for the water produced across different pH ranges.

It has partnered with several manufacturers and vendors based in Bengaluru and Coimbatore to supply customised components to build the product. The startup is looking to expand its operations from 2025 to other metro cities in the country and is ambitiously targeting a foreign location in Europe or the Middle East.

—Yashasvini Razdan

2 EMPOWERING SUSTAINABLE TRANSITIONS: TRANZITOR'S VISION FOR A BATTERY-POWERED FUTURE

The name Tranzitor draws its inspiration from one of the most important electronic components, the transistor. "Just as the transistor revolutionised electronics, Tranzitor aims to enable a transition towards a better alternative—battery technology," says Devesh Patel, Founder and CEO of Tranzitor

In today's world, as the urgency to address environmental challenges and create a sustainable future intensifies, the demand for deep tech startups that specialise in indigenous electric powertrain solutions has become paramount. Tranzitor is one such start-up that

is developing indigenous greener transportation solutions tailored to the Indian climate and road conditions. Their commitment to addressing the pressing issues of increasing pollution levels and combatting climate change is reflected in their range of indigenous solutions,

spanning from battery management systems (BMS) to motor controllers and beyond. With a keen focus on creating electric vehicle powertrain solutions suited for the unique road conditions in India, this startup hopes to make a significant impact in driving the transition towards